Open Flow Model

River Deep Mountain AI

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***Abstract*—The Open Flow Model is developed by River Deep Mountain AI (RDMAI), an innovation project funded by the Ofwat Innovation Fund. It aims to estimate daily mean river flow in ungauged catchments in Great Britain using a machine learning approach that leverages both static and dynamic data sources. This model enables scalable and timely estimation of flow metrics, bridging data gaps in remote rivers where gauged measurements are unavailable. In 80% of cases, it allows users to predict flows in ungauged catchments with an error margin of less than 0.8 m³/s.**

***Index Terms*—Water quality, *Flow* Estimation, machine learning, environmental monitoring, AI**

I. Introduction

Understanding river flow is essential for effective water resource management. River flow data, typically measured in cubic metres per second (m³/s), describes the volume of water moving through a waterbody at a given time. This information is critical for:

- Predicting floods
- Identifying, tracing and tracking pollution sources
- Managing ecosystems and water quality

However, gauged flow data is often sparse or entirely missing, especially in smaller or remote rivers. This is largely due to the high cost of installing and maintaining monitoring stations, leaving many catchments without direct observations. With a changing climate, understanding changes in flow is becoming increasingly important. To address these data gaps, hydrologists often rely on surrogate data from comparable catchments, deterministic hydrology models, or conduct expensive field campaigns. These approaches, while useful, are limited in scalability and timeliness.

II. Objective

Estimating daily mean river flow in ungauged catchments was the main objective of our work; however, this metric can be used to infer other key flow metrics, such as mean flow and Q95 (flow exceeded or equalled 95% of the time; usually taken as a metric of low flow). The model, and its associated performance metrics, presented in this report, represent the first iteration of our ML-model addressing the challenges mentioned above. Therefore, the results presented here are preliminary and should be considered as such.

A. Glossary

- **Catchments**: In this report, we refer to the area contributing to the flow of a river at a flow gauging station as a catchment. Consequently, the catchment of flow gauging station A will encompass the catchment of flow gauging station B if B is upstream of A on the same waterbody. In the development of our model, we have used data from 409 flow gauging stations, representing 409 catchments. This adheres to the use of catchments in similar literature ([1], [2], and [3]).
- **Static data**: Catchment attributes that rarely change, or change over longer periods of time, such as topography, land cover, and soil.
- **Dynamic data**: Data that changes daily or sub-daily, mainly covering weather data such as rainfall, temperature, radiation, etc.

B. Non-Stationarity

Due to the inherent non-stationarity of the challenges addressed in this project, it is important to continuously re-train and fine-tune the model to make sure it is up to date with the latest trends and variances of our climate and environment. The model was trained on publicly available data up to 2015. In the future, the model may not reflect updated scientific understandings, environmental conditions, or regulatory standards.

III. Data Sources

A. Primary Data Sources

1) **CAMELS-GB Dataset**
   - Provides catchment attributes and flow data across Great Britain (1970–2015).
   - Includes hydrological, topographical, soil, land cover, and human influence features.
2) **Hydrology Data API**
   - Offers flow flags and hydrological indicators across England.
   - Used for dynamic daily data such as precipitation, temperature, and radiation.
3) **Chalk Streams Dataset**
   - Supplied by Natural England.
   - Identifies unique geological catchments with distinct hydrological behaviour.
4) **Additional Variables**

- Includes derived features such as temporal indicators (e.g., sine/cosine of year and month) and lagged or rolling weather metrics.
- Feature engineering includes:
  - Lag periods: 1, 3, 7, and 30 days
  - Rolling windows: 3, 7, and 30 days

IV. Model Architecture

The Open Flow Model is designed to estimate **daily mean river flow (m³/s)** in ungauged catchments using a **hybrid neural network** that combines both **static** and **dynamic** data inputs.

A. Key Components

1) **TEMPORAL BRANCH:**
   - Uses a **Long Short-Term Memory (LSTM)** layer.
   - Processes **time-varying inputs** such as weather conditions.
   - Input sequences are **30-day sliding windows**, chosen to reflect catchment wetness and seasonal dynamics.
2) **STATIC BRANCH:**
   - Composed of **two dense layers** with decreasing size.
   - Encodes **static catchment attributes** such as land use, soil type, elevation, and hydrological indices.
3) **FUSION LAYER:**
   - Outputs from both branches are **concatenated**.
   - Passed to a final **dense layer** with either **exponential** or **softplus** activation to ensure non-negative flow predictions.

V. Train, Test, and Validation Strategies

A. Training and Testing Strategies

- **Per-Catchment Split:**
  - Each flow gauging station's time series is split 70% for training and 30% for testing.
  - This ensures the model is tested on unseen future data for each station, simulating real-world forecasting.
- **Time Windowing:**
  - Dynamic data (e.g., rainfall, temperature) is segmented into 30-day sliding windows.
  - This allows the LSTM to capture temporal dependencies and antecedent conditions such as soil moisture.
- **Static Feature Integration:**
  - Static attributes (e.g., topography, soil type, land cover) are encoded separately and concatenated with LSTM outputs.
  - This hybrid approach enriches predictions with both temporal and contextual information.

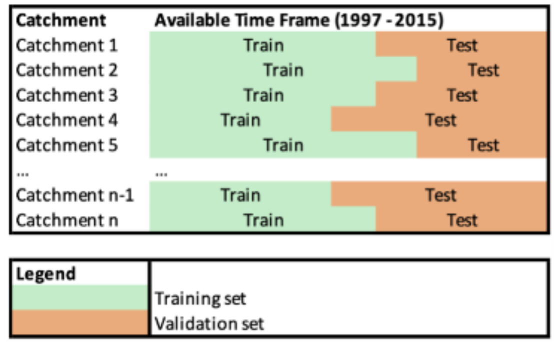

Fig. 1: Train Test split

B. Validation Strategy

- **Geographic Split:**
  - Catchments are split into train and validation sets based on geography (70/30), ensuring no overlap.
  - This prevents leakage and tests the model's ability to generalise to completely unseen catchments.
- **Stratified Sampling:**
  - To avoid statistical imbalance, target flow values are binned into five categories.
  - Stratification ensures similar distributions of flow characteristics across train and validation sets.
- **Multiple Validation Runs:**
  - 6 runs were conducted with different stratified or un-stratified splits. The first 3 unstratified runs were disregarded as train and validation were found to have differing distributions, hence the stratification.
  - Results on the remaining runs (04, 05, and 06) confirmed consistent performance and minimal bias across splits.

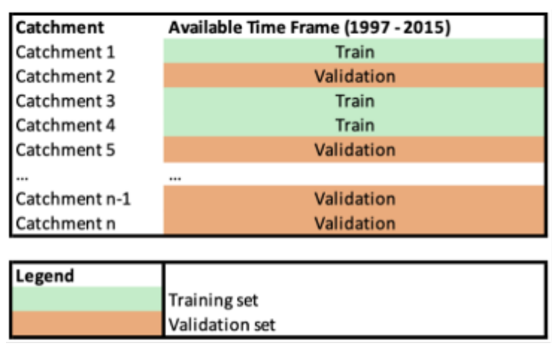

Fig. 2: Train validate split

C. Key Observations

- In testing, **MAE (Mean Absolute Error)** remained stable across runs (1.6–1.9 m³/s).
- In validation, **NSE (Nash-Sutcliffe Efficiency)** exceeded 0.9 globally in all runs, indicating strong predictive capability.
- About **80% of catchments** were high performing, while the remainder showed higher errors (often linked to chalk streams or extreme flow variability).

VI. Validation Results

The models demonstrated consistent performance across all runs, especially with respect to MAE. This showed that there were no significant statistical differences between the train and validation datasets for all runs. Please see the definitions sections for clarity on how we calculated the above metrics.

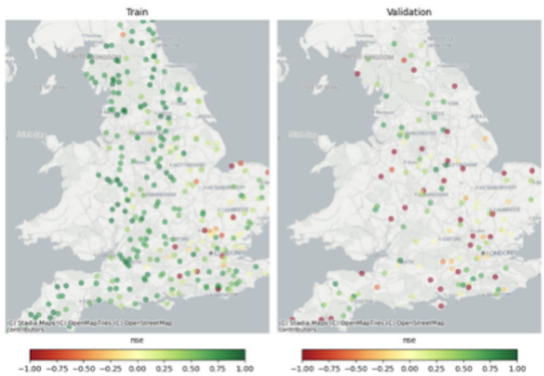

Fig. 3: Geographical representation of catchment NSE metrics for run 04.

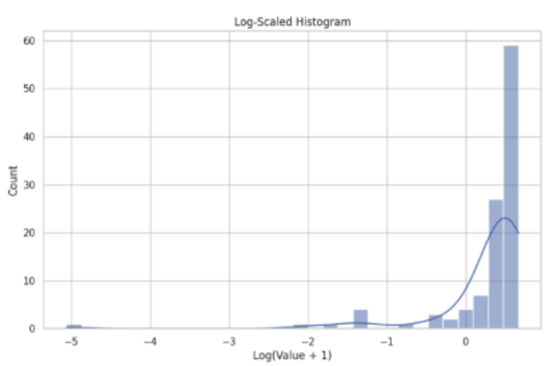

Fig. 4: Log-scaled histogram plot for NSE for run 04.

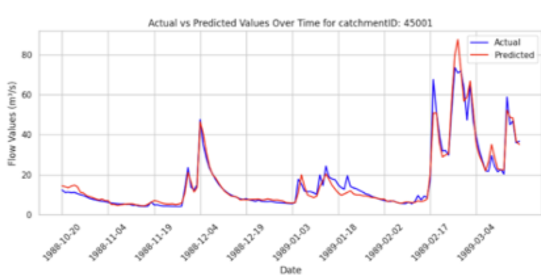

Fig. 5: Timeseries for actual vs preds for a high-performing catchment in run 04

VII. Evaluation Metrics

As a regression problem, the model was primarily trained using the **Mean Absolute Error (MAE)** as the loss function. MAE was selected for its straightforward interpretation and its focus on absolute accuracy. However, during both training and evaluation, several metrics were monitored to provide a comprehensive view of model performance:

- **MAE (Mean Absolute Error):** Measures the average magnitude of errors in absolute terms, providing a direct indication of how far predictions deviate from observed values.
- **MAPE (Mean Absolute Percentage Error):** Captures the average percentage error and is particularly useful for assessing relative performance across catchments with varying flow magnitudes.
- **NSE (Nash–Sutcliffe Efficiency):** Despite known limitations, this is a widely used metric in hydrological modelling. NSE evaluates the predictive power and accuracy of the model and ranges from $-\infty$ to 1, where 1 indicates a perfect prediction.

A. Catchment Level Aggregation

Given that the dataset includes time series from multiple flow gauging stations (each representing a different catchment), all metrics are also computed and reported as **catchment-averaged values**. This allows for the identification of spatial patterns in model performance and highlights regions where the model excels or underperforms. Errors are geographically mapped to visualise these patterns and support further analysis.

B. Error Concentation Analysis

Beyond average metrics, an **error concentration analysis** was conducted at the catchment level. This analysis identifies: Which catchments consistently exhibit higher errors and whether errors are localised to specific hydrological or geographical conditions, such as chalk streams or particular land-use types.

This layered evaluation approach provides a comprehensive understanding of model performance, both globally and locally, and helps guide future improvements in model generalisation. Visualisation through maps further enhances geographical reference and interpretability.

VIII. Performance Metrics

The performance of the Open Flow Model was rigorously evaluated using a suite of statistical metrics and visual analyses, ensuring a robust understanding of the model's strengths and limitations across diverse hydrological contexts.

A. Overview of Experimental Results

A series of experiments were conducted using slightly different datasets, with consistent loss functions and output layer structures to ensure comparability.

B. Key Findings

- **Consistency Across Experiments:** The model demonstrated consistent performance across all experimental runs, particularly with respect to MAE. This consistency indicates that the model is robust to minor variations in the dataset.

TABLE I: Validation results across three experimental runs

Run	MAE (Global)	Mean MAE (by catchment)	NSE (Global)	Mean NSE (by catchment)	Mean (Global)	Mean of means (by catchment)	SD (Global)	Mean of SDs (by catchment)
04	1.94	1.84	0.91	-0.19	6.96	6.31	25.89	7.04
05	1.57	1.43	0.93	-6.74	5.35	4.80	16.95	5.54
06	1.60	1.59	0.91	-0.75	5.09	5.07	17.65	6.06

- **Catchment-Level Insights:** Performance metrics were aggregated at the catchment level, revealing spatial patterns in model accuracy. Larger catchments tended to exhibit higher absolute errors, while smaller catchments showed more significant relative errors.
- **Error Concentration:** Analysis revealed that approximately 5% of catchments accounted for most errors. These "poor performing catchments" were further investigated to identify potential causes, such as unique hydrological characteristics or data limitations.
- **Geographical Mapping:** Errors and performance metrics were visualised geographically, enabling the identification of regions where the model performed particularly well or poorly. This spatial analysis supports targeted model improvements and resource allocation.
- **Catchment-Level Performance:** The distribution of errors at the catchment level indicated that, while the model generally overestimated flow, the magnitude of underestimation was greater when it occurred. This asymmetry was evident in the distribution tails and highlighted areas for further calibration. The distribution of catchment-level errors showed a general overestimation of flow, although differences between simulated and observed values were higher in those catchments where flow was underestimated. This asymmetry highlighted areas for further calibration.

C. Interpretations and Implications

The layered evaluation approach, combining global and local metrics, provided a nuanced understanding of model performance. The use of maps and visualisations facilitated the identification of spatial trends and outlier catchments, guiding future model development and calibration efforts. Overall, the performance metrics demonstrate that the Open Flow Model is capable of reliably estimating river flow across most catchments, with clear pathways identified for addressing areas of underperformance.

D. Benefit

The Open Flow Model offers several benefits for water companies and regulators in comparison to the current best-practice tool. These include:

- Being free and open to adapt; using open datasets.
- Being quicker and more cost-effective to run compared to available conceptual hydrological models.
- Providing estimates of flow considering human influences on flow rate.

IX. Conclusion

The Open Flow Model represents a significant advancement in the estimation of daily mean river flow in ungauged catchments, leveraging machine learning techniques and a diverse array of static and dynamic data sources. Through rigorous evaluation using multiple statistical metrics and validation strategies, the model has demonstrated strong generalisation across unseen catchments and consistent performance across various experimental runs.

While most catchments are modelled with high accuracy, a small subset of 'poor performing' catchments has been identified, highlighting areas for further investigation and targeted improvement. The layered evaluation approach—combining global and local metrics, spatial mapping, and error concentration analysis—has provided valuable insights into both the strengths and limitations of the model. Looking ahead, ongoing research has been recommended in the following areas: enhanced feature engineering, improved handling of challenging catchment types (such as chalk streams), and the integration of advanced interpretability tools such as SHAP value charts. These efforts aim to further increase the model's robustness, transparency, and applicability for water resource management and environmental decision-making.

Overall, the Open Flow Model provides a robust, scalable, and transparent framework for river flow estimation, supporting more informed management of water resources in the face of data scarcity and environmental change

X. Definitions

TABLE II: Summary of Metrics and their Descriptions

Metric	Description
MAE (Global)	Mean Absolute Error in daily average flow calculated across all data points globally.
MEAN MAE (by catchment)	Average of the MAE values computed individually for each catchment.
NSE (Global)	Nash-Sutcliffe Efficiency is calculated across all data points globally.
MEAN NSE (by catchment)	Average of the NSE values computed individually for each catchment.
Mean (Global)	The average (mean) daily average flow calculated across all data points globally.
MEAN of means (by catchment)	The mean daily average flow computed individually for each catchment.
SD (Global)	The Standard Deviation of all the data.
MEAN of SDs (by catchment)	The mean of the Standard Deviations of the catchments.

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Disclaimer

This document has been prepared by the RDMAI consortium and is intended for informational purposes only.

The model described in this document is the product of an ongoing research and development project. It represents an early iteration and its outputs are preliminary. The model is not intended for use in production environments, regulatory reporting, environmental monitoring, or any decision-making in relation to public health, public policy, or commercial matters without independent validation and the acceptance of full liability by the user.

All use of the model and its associated software is subject to the full terms set out in the Open Flow Model repository README, available at https://github.com/Cognizant-RDMAI/Open-Flow-Model. Readers seeking technical detail on model performance, limitations, and data sources should refer to the accompanying Model Report. The RDMAI consortium and its members accept no liability for any outcomes arising from use of or reliance on the model or its outputs.